

**Riphah International Colleges**  
**A Project of Riphah International University, Lahore**  
**Course Outline**

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                 |  |                      |                   |           |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|-------------------|-----------|
| Course Information             | Course Title                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Multimedia Production                                                                                                                                                                                           |  |                      |                   |           |
|                                | Course ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CS-2233                                                                                                                                                                                                         |  | Course Type          | Major             |           |
|                                | Credit hours                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3 (3-0)                                                                                                                                                                                                         |  | Hours per week (C-L) | 3 (3-0)           |           |
|                                | Programs                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ADP IT-Management                                                                                                                                                                                               |  | Preferred Semester   |                   |           |
|                                | Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 <sup>th</sup> Semester                                                                                                                                                                                        |  | Version              | 2.0               |           |
| Course Description             | This is a practicum course applying the pragmatic approach to give you the opportunity to apply technical knowledge in In Digital media production. In today’s world the digital media production is of immense importance that require the human resource that can meet the challenges of content market. This course will provide all the aspect of media production from tools to theoretical knowledge that can be applied to both social media and other platforms. |                                                                                                                                                                                                                 |  |                      |                   |           |
| Course Objectives (CO)         | The objective of this course is to enable students to understand;                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                 |  |                      |                   |           |
|                                | No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Objective                                                                                                                                                                                                       |  |                      | Relation with PEO |           |
|                                | CO1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Learn the history of multimedia and how human brains process information through sight and sound using the Dual Coding and Cognitive Load theories.                                                             |  |                      |                   |           |
|                                | CO2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Develop hands-on skills in professional software including <b>Photoshop/Figma</b> for images, <b>Audacity</b> for sound, and <b>Premiere Pro/CapCut</b> for high-quality video production.                      |  |                      |                   |           |
|                                | CO3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Learn to design interactive products, simulations, and basic animations using <b>Scratch</b> to keep audiences engaged.                                                                                         |  |                      |                   |           |
|                                | CO4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Practice the complete "real-world" <b>process of media creation</b> : from initial planning and budgeting (costing) to designing, producing, and finally publishing content for social media and web platforms. |  |                      |                   |           |
|                                | CO5.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Gain the ability to look at existing websites or videos and identify if they are easy to use (navigation) and effective for learning (instructional design)                                                     |  |                      |                   |           |
| Course Learning Outcomes (CLO) | At the end of this course students will be able to;                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                 |  |                      |                   |           |
|                                | No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Outcome                                                                                                                                                                                                         |  | Relation with PLO    | BT Level          | PLO Level |
|                                | CLO1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Describe how people learn from words and pictures by identifying key rules like the Dual Coding and Cognitive Load theories.                                                                                    |  | PLO-1                | C2                |           |
|                                | CLO2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Create and edit high-quality images, audio, and videos using professional tools like <b>Photoshop, Audacity, and Premiere Pro/CapCut</b> .                                                                      |  | PLO-2                | P3                |           |
|                                | CLO3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Develop simple interactive projects and animations using tools like <b>Scratch</b> to show how users can engage with digital media.                                                                             |  | PLO-3                | C3                |           |
|                                | CLO4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Analyze and critique existing videos or websites to see if they follow good design rules and are easy for people to use.                                                                                        |  | PLO-4                | C4                |           |
|                                | CLO5.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Organize a media project from start to finish, including planning the idea, estimating costs, and saving files in the right formats for social media or the web.                                                |  | PLO-3                | C3                |           |
| Lecture type                   | Lectures & Practical Demonstrations                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                 |  |                      |                   |           |
| Prerequisites                  | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                 |  |                      |                   |           |
| Follow up Courses              | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                 |  |                      |                   |           |

**TEXTBOOKS AND REFERENCE BOOKS:**

| Textbook | Title | Editi | Authors | Publisher | Year | ISBN |
|----------|-------|-------|---------|-----------|------|------|
|----------|-------|-------|---------|-----------|------|------|

|                                      |                                                                              |                                                                                      |                                 |                                |      |               |
|--------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------|--------------------------------|------|---------------|
|                                      |                                                                              | on                                                                                   |                                 |                                |      |               |
|                                      | <a href="#">The Cambridge Handbook of Multimedia Learning</a>                |                                                                                      | Mayer, R.E.                     | Cambridge University Press.    | 2005 |               |
| Reference Books & Reference Material | <a href="#">Multimedia learning.</a>                                         |                                                                                      | Mayer, R.E.                     | Cambridge University Press.    | 2001 |               |
|                                      | <a href="#">Instructional media and technologies for learning</a>            |                                                                                      | Upper Saddle & River, NJ        | Prentice-Hall                  | 1999 |               |
|                                      | <a href="#">Multimedia for Learning: Methods and Development</a>             |                                                                                      | Alessi, S. M., & Trollip, S. R. | Boston, MA: Allyn & Bacon, Inc | 2001 | 9780205276912 |
|                                      | <a href="#">Adobe Fireworks CS5 Classroom in a Book: Adobe Training book</a> |                                                                                      | Jim Babbage                     | Adobe Press                    |      |               |
|                                      | Audacity. The Free, Cross-Platform Sound Editor                              | <a href="http://audacity.sourceforge.net">http://audacity.sourceforge.net</a>        |                                 |                                |      |               |
|                                      | Scratch Tool Online Tool                                                     | <a href="http://scratch.mit.edu/">http://scratch.mit.edu/</a>                        |                                 |                                |      |               |
|                                      | <b>Books QR-Code</b>                                                         |    |                                 |                                |      |               |
| Tools and Technologies               |                                                                              | Audacity, Windows Movie Maker, Scratch, Premiere Pro/CapCut and Photoshop/Figma etc. |                                 |                                |      |               |
| Design skills / techniques           |                                                                              |                                                                                      |                                 |                                |      |               |
| Helping Material                     |                                                                              | <a href="#">CS-subject-Slides.url</a>                                                |                                 |                                |      |               |

## Course Outline (Multimedia Production)

### Course Contents

| Week No. | Topic                                                       | Lecture No. | Lecture Contents                                                                                                                                                                                                                                                                                                                       | Relation with CLO | Lecture Material                                                                                                            | Class Activity              | Tasks |
|----------|-------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------|
| W1.      | Overview and Introduction                                   | L1          | <ul style="list-style-type: none"> <li>Course Overview and Study Scheme Introduction to Multimedia, Hypermedia and different related key terms</li> </ul>                                                                                                                                                                              | CLO-1             | <ul style="list-style-type: none"> <li>Mayer (2021): Ch. 1 (The Promise of Multimedia Learning); Heinich: Ch. 1.</li> </ul> |                             |       |
|          |                                                             | L2          | <ul style="list-style-type: none"> <li>Brief history of multimedia</li> <li>Benefits and examples of multimedia, hypermedia, hypertext.</li> </ul>                                                                                                                                                                                     |                   |                                                                                                                             |                             |       |
| W2.      | Multimedia Learning Theories                                | L3 & L4     | <ul style="list-style-type: none"> <li>Theories of Multimedia Learning <ul style="list-style-type: none"> <li>Cognitive Theory of Multimedia</li> <li>Dual Coding Theory</li> </ul> </li> </ul>                                                                                                                                        | CLO-1             | <ul style="list-style-type: none"> <li>Mayer (2021): Ch. 3 (Cognitive Theory of Multimedia Learning).</li> </ul>            | Multimedia Project Briefing |       |
| W3.      | Implication of Theories and Integrated Models               | L5 & L6     | <ul style="list-style-type: none"> <li>Implications of Cognitive Load Theory to Multimedia Learning</li> <li>Integrated Models of Text and Picture Comprehension</li> </ul>                                                                                                                                                            | CLO-1, CLO-2      | <ul style="list-style-type: none"> <li>Mayer (2021): Ch. 4; Heinich: Ch. 3 (Visual Principles).</li> </ul>                  | Assignment 1                |       |
|          |                                                             |             | <ul style="list-style-type: none"> <li>The Four-Component Instructional Design Model</li> </ul>                                                                                                                                                                                                                                        |                   |                                                                                                                             |                             |       |
| W4.      | Graphics Functions & Multimedia Principles Overview         | L7          | <ul style="list-style-type: none"> <li>Functions of Graphics</li> <li>Multimedia Learning Improvement techniques?</li> </ul>                                                                                                                                                                                                           | CLO-2             | <ul style="list-style-type: none"> <li>Heinich: Ch. 4 (Design Principles); Non-Designer's Design Book.</li> </ul>           | Quiz 1                      |       |
|          |                                                             | L8          |                                                                                                                                                                                                                                                                                                                                        |                   |                                                                                                                             |                             |       |
| W5.      | Multimedia Principles & Visual Design Principles            | L9 & L10    | <ul style="list-style-type: none"> <li><b>Multimedia Principles in Environments for Complex Learning</b></li> <li><b>Visual Design Principles:</b> Balance, Harmony, Closure, Proximity, Contrast, Color, Alignment, Emphasis</li> </ul>                                                                                               | CLO-1, CLO-4      | <ul style="list-style-type: none"> <li>Mayer (2021): Ch. 5 (Signaling) &amp; Ch. 10 (Segmenting).</li> </ul>                | Assignment 2                |       |
| W6.      | Principles & Techniques for Reducing Extraneous Processing  | L11 & L12   | <ul style="list-style-type: none"> <li><b>Principles &amp; Techniques for Reducing Extraneous Processing</b> <ul style="list-style-type: none"> <li>Coherence principle</li> <li>Signaling principle</li> <li>Redundancy principle</li> <li>Spatial contiguity principle</li> <li>Temporal contiguity principle</li> </ul> </li> </ul> | CLO-1, CLO-4      | <ul style="list-style-type: none"> <li>Mayer (2021): Ch. 6 (Coherence) &amp; Ch. 8-9 (Contiguity).</li> </ul>               | Quiz 2                      |       |
| W7.      | Principles & Techniques for Fostering Generative Processing | L13 & L14   | <ul style="list-style-type: none"> <li><b>Principles &amp; Techniques for Fostering Generative Processing</b> <ul style="list-style-type: none"> <li>Segmenting principle</li> <li>Pre-training principle</li> <li>Modality principle</li> </ul> </li> </ul>                                                                           | CLO-1             | <ul style="list-style-type: none"> <li>Mayer (2021): Ch. 13 (Personalization) &amp; Ch. 14 (Voice).</li> </ul>              |                             |       |

| Week No. | Topic                                                                | Lecture No. | Lecture Contents                                                                                                                                                                                                                                                                      | Relation with CLO | Lecture Material                                                                                                                                                                   | Class Activity | Tasks |
|----------|----------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------|
| W8.      | Principles & Techniques for Managing Essential Processing            | L15 & L16   | <ul style="list-style-type: none"> <li>Principles &amp; Techniques for Managing Essential Processing               <ul style="list-style-type: none"> <li>Personalization principle</li> <li>Voice principle</li> </ul> </li> </ul>                                                   | CLO-2             | <ul style="list-style-type: none"> <li>Mayer (2021): Ch. 13 (Personalization) &amp; Ch. 14 (Voice).</li> </ul>                                                                     |                |       |
| W9.      | Mid Term Exam Week                                                   | L17 & L18   | Mid Term Examination (CLO-1, 2 and 4)                                                                                                                                                                                                                                                 |                   |                                                                                                                                                                                    |                |       |
| W10.     | Introduction & Applications of Multimedia Tools                      | L19         | <ul style="list-style-type: none"> <li>Introduction to Multimedia Tools &amp; Trends</li> </ul>                                                                                                                                                                                       | CLO-2             | <ul style="list-style-type: none"> <li>Heinich: Ch. 7 (Audio); Audacity User Manual.,</li> <li>Ch.8 (Video Production); Premiere Pro/CapCut Guides.</li> </ul>                     |                |       |
|          |                                                                      | L20         | <ul style="list-style-type: none"> <li>Applications of different Multimedia Tools (Audacity, Windows Movie Maker, Scratch, Premiere Pro/CapCut and Photoshop/Figma etc.)</li> </ul>                                                                                                   |                   |                                                                                                                                                                                    |                |       |
| W11.     | Creating Multimedia using Multimedia Tools                           | L21 & L22   | <ul style="list-style-type: none"> <li>Creating Multimedia Text, Sound, Images, Video and animation using Multimedia Tools</li> </ul>                                                                                                                                                 | CLO-2             | <ul style="list-style-type: none"> <li>Mayer (2021): Ch. 15 (Social Cues in Multimedia Learning).</li> </ul>                                                                       | Assignment 3   |       |
| W12.     | Principles of Stack Design & Mayer's Multimedia Learning             | L23         | <ul style="list-style-type: none"> <li>Principles of Stack Design in Multimedia Learning</li> </ul>                                                                                                                                                                                   | CLO-3, CLO-4      | <ul style="list-style-type: none"> <li>Alessi &amp; Trollip: Ch. 2 (Interactivity &amp; Navigation),</li> <li>Mayer (2005) Handbook: Section II (Principles of Design).</li> </ul> | Quiz 3         |       |
|          |                                                                      | L24         | <ul style="list-style-type: none"> <li>Mayer's Multimedia Learning Principles in detail</li> <li>Relation with Stack Design Principles and Mayers Multimedia Learning Principles</li> </ul>                                                                                           |                   |                                                                                                                                                                                    |                |       |
| W13.     | Publishing Steps for Multimedia Projects with different file formats | L25         | <ul style="list-style-type: none"> <li>Publishing Steps for Multimedia Projects               <ul style="list-style-type: none"> <li>Optimization and Export Methods</li> <li>Platform Specific Delivery formats and Methods</li> <li>Post Publishing Workflow</li> </ul> </li> </ul> | CLO-3, CLO-1      | <ul style="list-style-type: none"> <li>Alessi &amp; Trollip: Ch. 12 (Quality &amp; Distribution).</li> </ul>                                                                       |                |       |
|          |                                                                      | L26         | <ul style="list-style-type: none"> <li>Different Multimedia Publishing file formats with characteristics</li> </ul>                                                                                                                                                                   |                   |                                                                                                                                                                                    |                |       |
| W14.     | Multimedia Delivery Process:                                         | L27 & L28   | <ul style="list-style-type: none"> <li>Multimedia Delivery Process:               <ul style="list-style-type: none"> <li>Planning: The Foundation</li> <li>Costing: Budgeting and Resources</li> <li>Designing the Blueprint</li> <li>Producing Deliverables</li> </ul> </li> </ul>   | CLO-5             | <ul style="list-style-type: none"> <li>Heinich: Ch. 10 (Designing &amp; Producing Media).</li> </ul>                                                                               | Assignment 4   |       |

| Week No. | Topic                          | Lecture No. | Lecture Contents                                                                                                                                                                                                                                                                                                                                                                                                                            | Relation with CLO | Lecture Material                                                                                                      | Class Activity | Tasks |
|----------|--------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------|----------------|-------|
| W15.     | Different Multimedia Learning  | L29 & L30   | <ul style="list-style-type: none"> <li>• Multimedia Learning in <b>Advanced Computer-Based Contexts</b></li> <li>• Multimedia Learning with <b>Animated Pedagogical Agents</b></li> <li>• Multimedia Learning in <b>Virtual Reality</b></li> <li>• Multimedia Learning with <b>Games, Simulations, and Micro worlds</b></li> <li>• Multimedia Learning with <b>Hypermedia</b></li> <li>• Multimedia Learning in <b>E-Courses</b></li> </ul> | CLO-5             | <ul style="list-style-type: none"> <li>• Mayer (2021): Ch. 16 (VR), Ch. 17 (Agents), Ch. 18 (Games).</li> </ul>       | Quiz 4         |       |
| W16.     | Advanced Multimedia Principles | L31 & L32   | <ul style="list-style-type: none"> <li>• <b>Advanced Multimedia Principles for</b> <ul style="list-style-type: none"> <li>○ Guided-discovery</li> <li>○ Workout example</li> <li>○ Collaboration</li> <li>○ Self-explanation</li> <li>○ Animation and interactivity</li> <li>○ Navigation</li> <li>○ Site map</li> <li>○ Prior knowledge</li> <li>○ Cognitive aging</li> </ul> </li> </ul>                                                  | CLO-3, CLO-4      | <ul style="list-style-type: none"> <li>• Mayer (2021): Ch. 19 (Cognitive Aging &amp; Advanced Principles).</li> </ul> |                |       |
| W17.     | Presentation & Revision        | L33 & L34   | <ul style="list-style-type: none"> <li>• Multimedia Project <b>Presentations</b> and <b>Revision</b></li> </ul>                                                                                                                                                                                                                                                                                                                             |                   |                                                                                                                       | Presentations  |       |
| W18.     |                                |             | <ul style="list-style-type: none"> <li>• Final Term Exam</li> </ul>                                                                                                                                                                                                                                                                                                                                                                         |                   |                                                                                                                       |                |       |